Chapter 1

Charge

If you rub a balloon against your hair, then place it next to a wall, it will stick. This is because it stole some electrons from your hair, and now the ballon has slightly more electrons than protons. We say that it has gotten an *electrical charge*, or an imbalance between protons and electrons. In this case, the balloon has a negative electrical charge. Objects with slightly more protons than electrons have a positive charge. Consequently, your hair now has a positive charge. Note that the differences are equal but opposite. While *charge* can, *net charge cannot be created or destroyed*.

An object with a negative charge and an object with a positive charge will be *attracted* to each other. Two objects with the same charge will be *repelled* by each other.

This charge, referred to as *elementary charge* is represented by the base unit e (not to be confused with the e^x function in math disciplines). This charge value e cannot be a decimal since it is a unit, just like counting coins. You cannot have a fractional amount of coins,it must be countable.

In this way, we call elementary charge *quantized*. This was shown back in Because it is quantized, we represent the charge of a particle as q. The SI (International System of Units) measures charge in in *coulombs*. The charge of a single proton is about 1.6×10^{-19} coulombs.

Two objects with charges are bound to either attract (pull) or repel (push) on each other. This is a type of force referred to as *electric force*.

Coulomb's Law

If two objects with charge q_1 and q_2 (in coulombs) are r meters from each other, the force of attraction or repulsion (electric force) is given by

$$F = K\frac{q_1 q_2}{r^2} \tag{1.1}$$

where F is in newtons and K is Coulomb's constant: about 8.988×10^9. You may use $9 \times 10^9\,\mathrm{N \cdot m^2/C}$ in your calculations.

This k value comes from $\frac{1}{4\pi\epsilon_0}$, a *relationship with the permittivity of free space*, ϵ_0.

You may see this as F_e, F_q, or F_c, but they all refer to the electric force and the standard is F_e.

Note how similar this looks to Newton's Law of Universal Gravitation:

$$F_g = G\frac{m_p m_p}{r^2}$$

Both laws follow an inverse-square relationship, meaning the force decreases with the square of the distance. However, gravity is always attractive, while electric force can be either attractive or repulsive depending on the signs of the charges (q_1 and q_2).

If this force is negative, it represents the idea that the two given particles are attracted to each other, as this would mean one of the particles must have a negative charge.

If this force is positive, it would mean there is a push or repelling force on each of the particles.

What if this force is zero? This would only result if one or both particles have zero charge. This would mean that they do not have an electric force, $F_E = 0$.

Exercise 1 Coulomb's Law

Two balloons are charged with an identical quantity and type of charge: -5×10^{-9} coulombs. They are held apart at a separation distance of 12 cm. Determine the magnitude of the electrical force of repulsion between them.

Working Space

Answer on Page 9

At this point, you might ask "If the wall has zero charge, why is the balloon attracted to it?" The answer: the electrons in the wall move away from the balloon, polarizing the atoms. The negative charge on the balloon pushes electrons away from itself, so the surface of the wall gets a mild positive charge. The negative charge on the balloon is attracted more to the positive charge on the surface of the wall than the negative charge on the inside, thus the balloon sticks. There's some weirdness going on with conductors and insulators here, but we will get to that later.

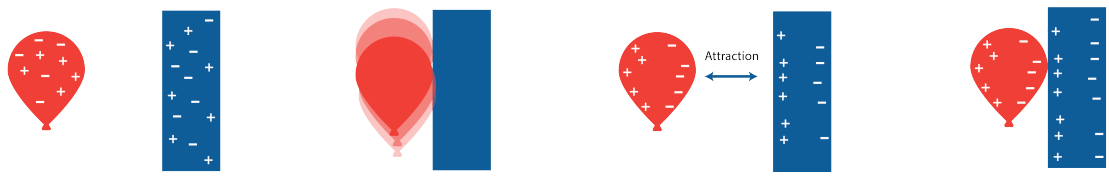
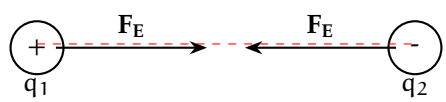


Figure 1.1: The balloon is attracted to the wall as the balloons positive charge tries to rejoin with the wall's negative charge.

1.0.1 Point Charge

The wall and the balloon all contain a bunch of tiny particles which we can refer to as **point charges**. A point charge is an electric charge that is treated as a concentrated single point in space, with no physical size or volume. Similar to point masses, we use point charges to simulate effects of charge on other objects or points.

This charge acts parallel to the secant line that touches both point charges only once.



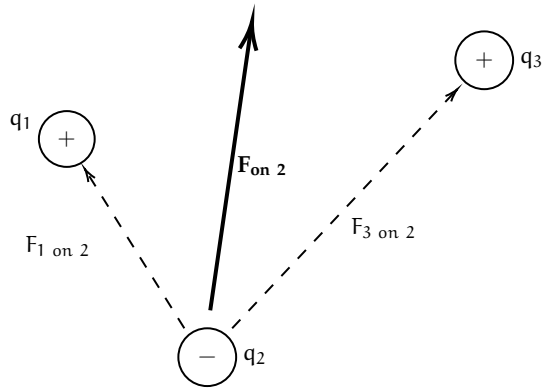
1.0.2 Superposition

What if there are more just one charges acting on a given point charge? Coulomb's law gives the force between two point charges, but in a system with three or more charges we apply the **principle of superposition**[1].

The principle of superposition states that the total electric force on a charge is the vector sum of the individual forces exerted on it by all other charges.[2]

[1]Superposition is really a linear algebra topic, and not something that this textbook will cover, but it has so many real life applications and we recommend you do your own research! See the wiki page for more: https://en.wikipedia.org/wiki/Superposition_principle

[2]This is similar how we calculate tensions when there are multiple tensions (or just forces in general) on an object. The net force is the sum of all the forces after vector addition is applied.



Exercise 2 Finding net charge on a point charges

Working Space

This question is taken from the AP Physics C 19th edition practice workbook.

Consider four equal, positive point charges that are situated at the vertices of a square. Find the net electric force on a negative point charge placed at the square's center.

Hint: draw a diagram to help you

Answer on Page 9

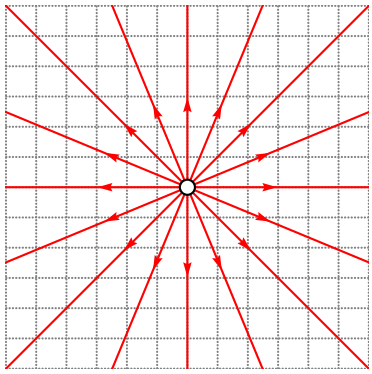
1.1 Electric Field

Recall that, when we talked about Gravitational Fields, we defined them as fields of infinitesimal points that each have a magnitude of force associated with them. This field revolves around the entire 2D circle or 3D sphere (generally point masses, charges, fields, etc are spherical).

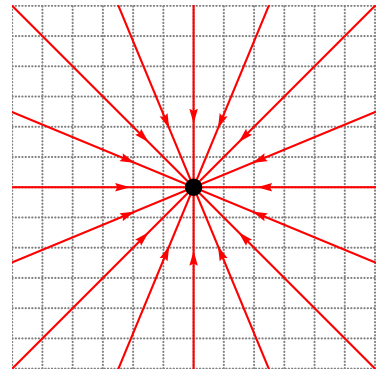
Similar to how the gravitational field extends out from the center of a given point mass, an *electric field* of a point charge has (imaginary) vectors pointing out of it. However, the direction of the vectors[3] (and electric force) depends on the the charge of the point

[3] You may hear the terms electric field lines and electric field vectors used interchangibly

charge. A negatively charged point mass has force lines pointing *towards* it, and a positively charged point mass has force lines pointing *away* from it (1.2a). In short, they always originate from positive charges and approach negative charges.



(a) A positively (open circle) charged electric field with corresponding electric field lines.



(b) A negatively (filled circle) charged electric field with corresponding electric field lines.

Figure 1.2: Two electric fields with different charges.

Imagine that there is a charge Q and a tiny particle test point q_t. Moving q_t, the test particle around, away from, further, and closer to the electric field of Q changes the force that Q feels. The force changes based on where the test particle q_t is. Recall that force is a vector, and the electric field is also a vector, so we can create a resulting formula:

$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}_e}{q_t} \tag{1.2}$$

What are the *units* of an electric field? Well, by this equation, $\vec{\mathbf{F}}_e$ is a force which we know is in *Newtons* (N), and the test particle q is in Coulombs (C). The resulting units are $\mathrm{N\,C^{-1}}$, or Newton's per Coulomb. Note that this is equivalent to $1\,\mathrm{V\,m^{-1}}$, or one Volt per meter[4]. This is how electric field forces are converted to volts within batteries or circuits!

It is important to note that we can simplify this this equation by substituting Coulomb's law:

$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}_e}{q_t} = \frac{\frac{kq_1q_t}{r^2}}{q_t} = \frac{kQ}{r^2} \tag{1.3}$$

where Q is the electric field's source charge.

For postive charges Q, the force vectors get weaker the farther from the charge (visually, the force lines get shorter). In fact, it decreases by a factor of $\frac{1}{r^2}$. Think to yourself where this factor comes from. (Hint: refer back to Coulomb's law (1.1))

[4]Equivallently, one $\mathrm{kV\,cm^{-1}}$, or kilovolt per centimeter is equal to $10^{-5} \times 1\mathrm{N\,C^{-1}}$. To reverse the operation, multiply the kilovolts per centimeter by 10^5 to get the resulting Newtons per Coulomb

Exercise 3 Point Charges and Electric Fields

This problem is adapted from an Example IB Exam. Two isolated point charges, X of charge +Q and Y of charge +2Q, are seperated by distance 3d. Point P is distance d from X and distance 2d from Y.

What is the net electric field strength at P? Are the forces from each charge equal?

1. 0, equal forces
2. $\frac{2kQ}{d^2}$, equal forces
3. $\frac{kQ}{2d^2}$, equal forces
4. $\frac{2kQ}{1.5d^2}$, equal forces

Working Space

Answer on Page 9

The density of the electric force lines at a given point determines the strength of the force at the point. Locations where field lines are more dense are much stronger than areas where the lines are sparse. This goes back to the idea of the force strength of an electric field.

1.2 Lightning

A cloud is a cluster of water droplets and ice particles. These droplets and ice particles are always moving up and down through the cloud. Recall that heat rises, so water is often trying to get higher in the cloud. In this process, electrons get stripped off and end up on the water droplets at the bottom of the cloud (water droplets collect at the bottom because they are denser). The air between the droplets is a pretty good insulator, which means the electrons are reluctant to jump anywhere. However, eventually, the charge gets so strong that even the insulating properties of the air is not enough to prevent the jump to earth, causing lightning. *Storms* are dense in clouds, so there is constant jumping of the electrons, which is why storms are prone to produing so much lightning. These electrons

really want to jump to earth, and when given enough time and ability to bypass the air's density. The electrons energy is visible as light and heat, and so we see a spark in the shape of a jagged line hitting a single point, the most conductive point on earth closest to the cloud. A great deal of lightning moves within a cloud or between clouds. However, sometimes it jumps to the earth. These bolts of lightning vary in the amount of electrons they carry, but the average is about 15 coulombs.

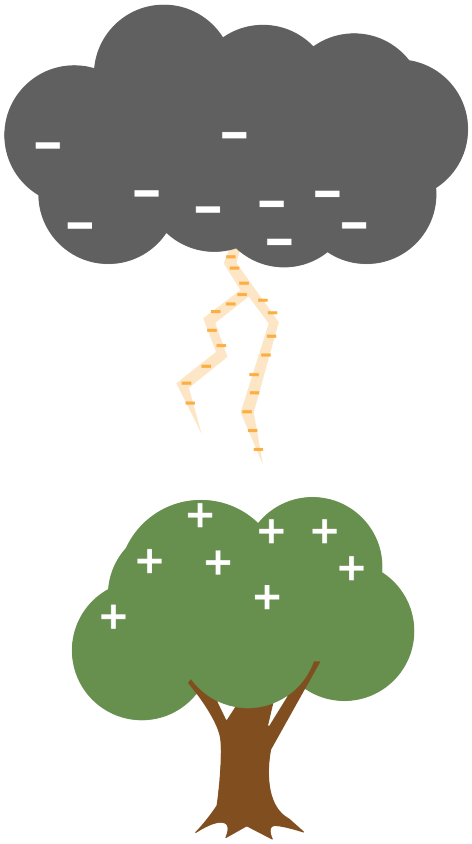


Figure 1.3: A difference of charge causes a spark of lightning to generate and transfer.

Thunder occurs because the electrons heat the air they pass through, causing the air to expand suddenly. Tthe resulting shockwave is the sound we know as thunder.

1.3 But...

This idea that opposite charges attract creates some heavy questions that you do not yet have the tools to work with. So to these questions, the answer is basically "Don't ask that yet!"

However, you probably have these questions, so we will point you in the direction of the answers.

The first is "In any atom bigger than hydrogen, there are multiple protons in the nucleus. Why don't the protons push each other out of the nucleus?"

We aren't ready to talk about it, but there is a force called *the strong nuclear force*, which pulls the protons and neutrons in the nucleus of the atom toward each other. At very, very small distances, it is strong enough to overpower the repulsive force due to the protons' charges.

Another question is "Why do the electrons whiz around in a cloud so far from the nucleus of the atom? Negatively charged electrons should cling to the protons in the center, right?"

We aren't ready to talk about this either, but quantum mechanics tells us that electrons like to live in a certain specific energy level. Hugging protons isn't one of those levels.

This is a draft chapter from the Kontinua Project. Please see our website (https://kontinua.org/) for more details.

Appendix A

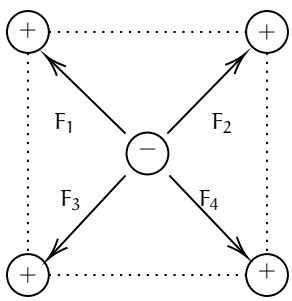
Answers to Exercises

Answer to Exercise 1 (on page 2)

$$F = K\frac{|q_1 q_2|}{r^2} = (8.988 \times 10^9)\frac{(-5 \times 10^{-9})(-5 \times 10^{-9})}{0.12^2} = \frac{224.7 \times 10^{-9}}{0.0144} = 15.6 \times 10^{-6}$$

15.6 micronewtons.

Answer to Exercise 2 (on page 4)



The attractive forces of F_1 and F_3 cancel each other out. Similarly, the attractive forces of F_2 and F_4 also cancel each other out. The net charge on the negative point charge is zero.

Answer to Exercise 3 (on page 6)

We noted above that the electric field caused by a point charge is defined as $E = \frac{kq}{r^2}$. We need to find the electric fields for each charge and then subtract their strengths.

Inputting known variables as constants for X from point P:

$$E = \frac{kQ}{d^2}$$

And doing the same for Y:

$$E = \frac{2kQ}{(2d)^2} = \frac{kQ}{2d^2}$$

Because the fields are in opposite directions, we subtract the smaller magnitude from the larger one. Let's define *to the right* as the positive direction:

$$E_{net} = E_X - E_Y \implies \frac{kQ}{d^2} - \frac{kQ}{2d^2}$$

To subtract, we find a common denominator:

$$E_{net} = \frac{2kQ}{2d^2} - \frac{kQ}{2d^2} = \frac{kQ}{2d^2}$$

The correct answer is C.

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